



University of Asia Pacific
Department of Computer Science and
Engineering

Design and Testing Of VLSI Lab
(CSE 458)

Submitted By

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Section: C-2

Semester: 4-2

Submitted To

Rashik Rahman

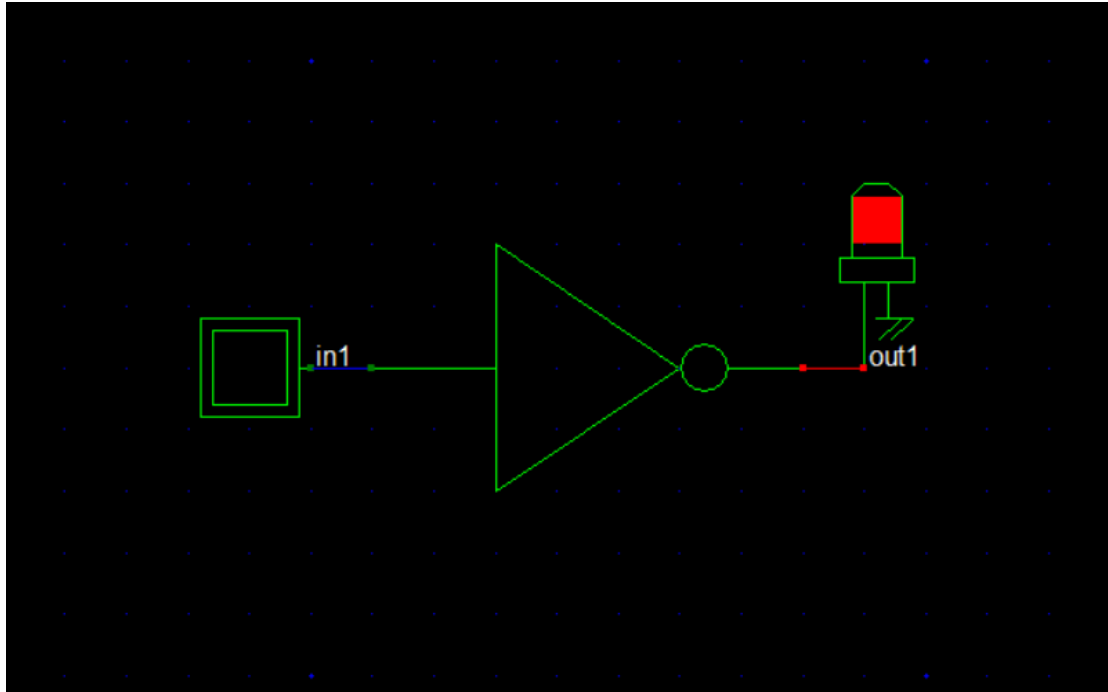
Lecturer

Department of Computer Science
and Engineering

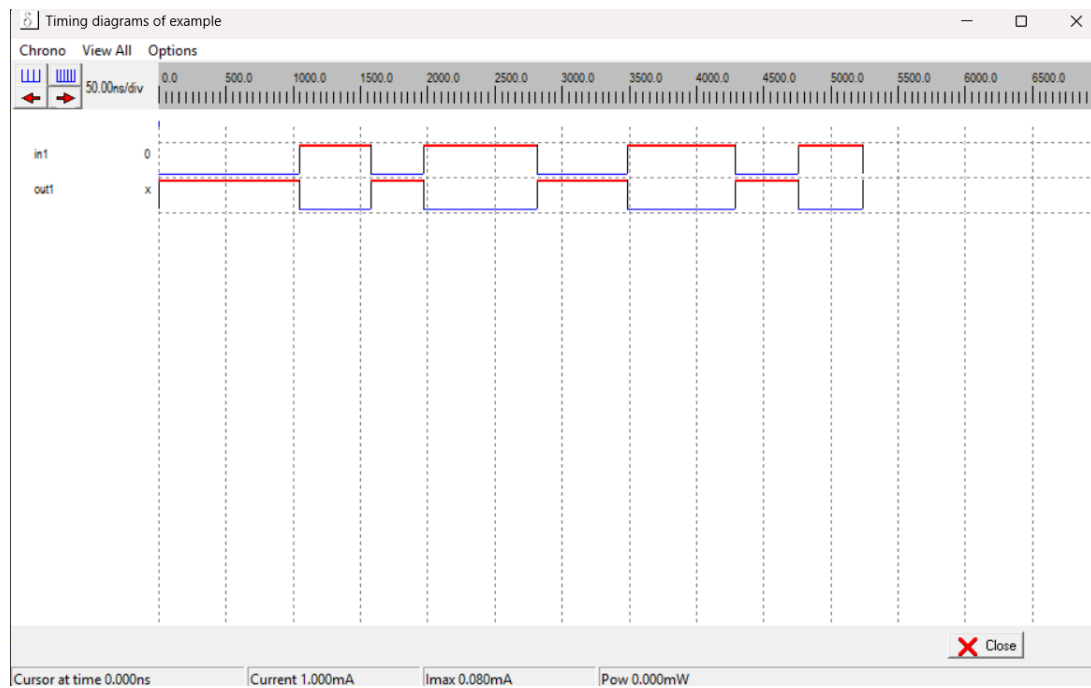
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Circuit 01 - Inverter

Circuit -

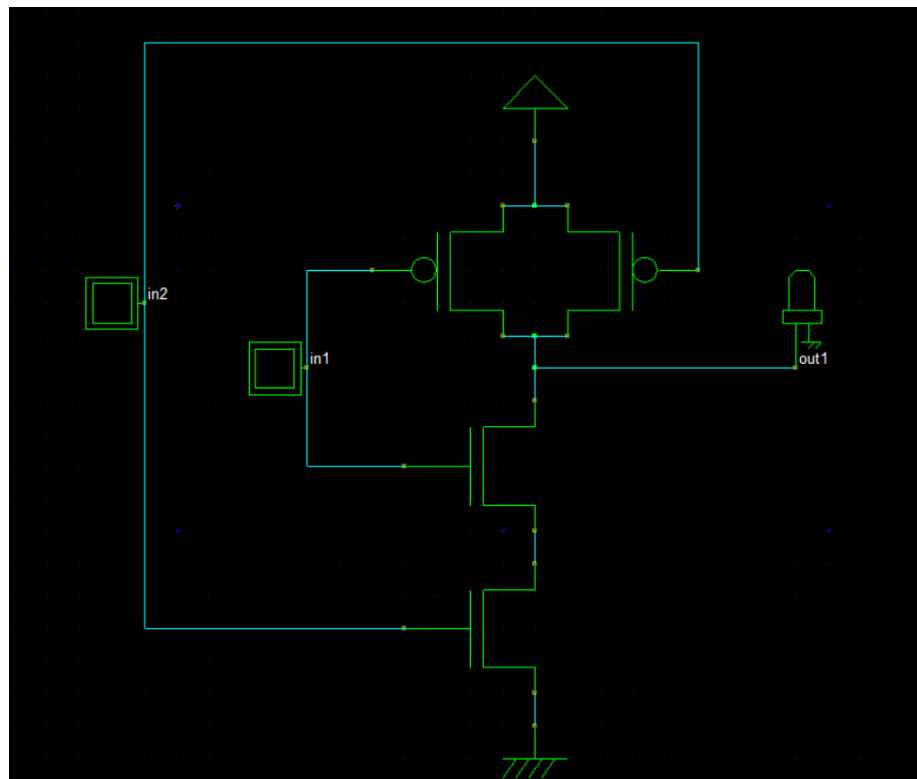


Timing Diagram -

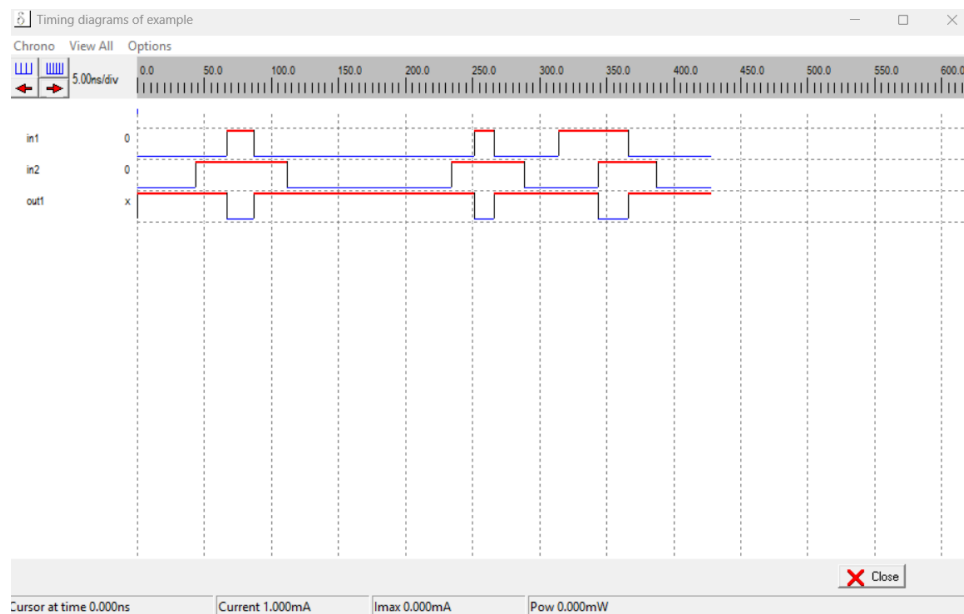


Circuit 02 - Nand Gate

Circuit -

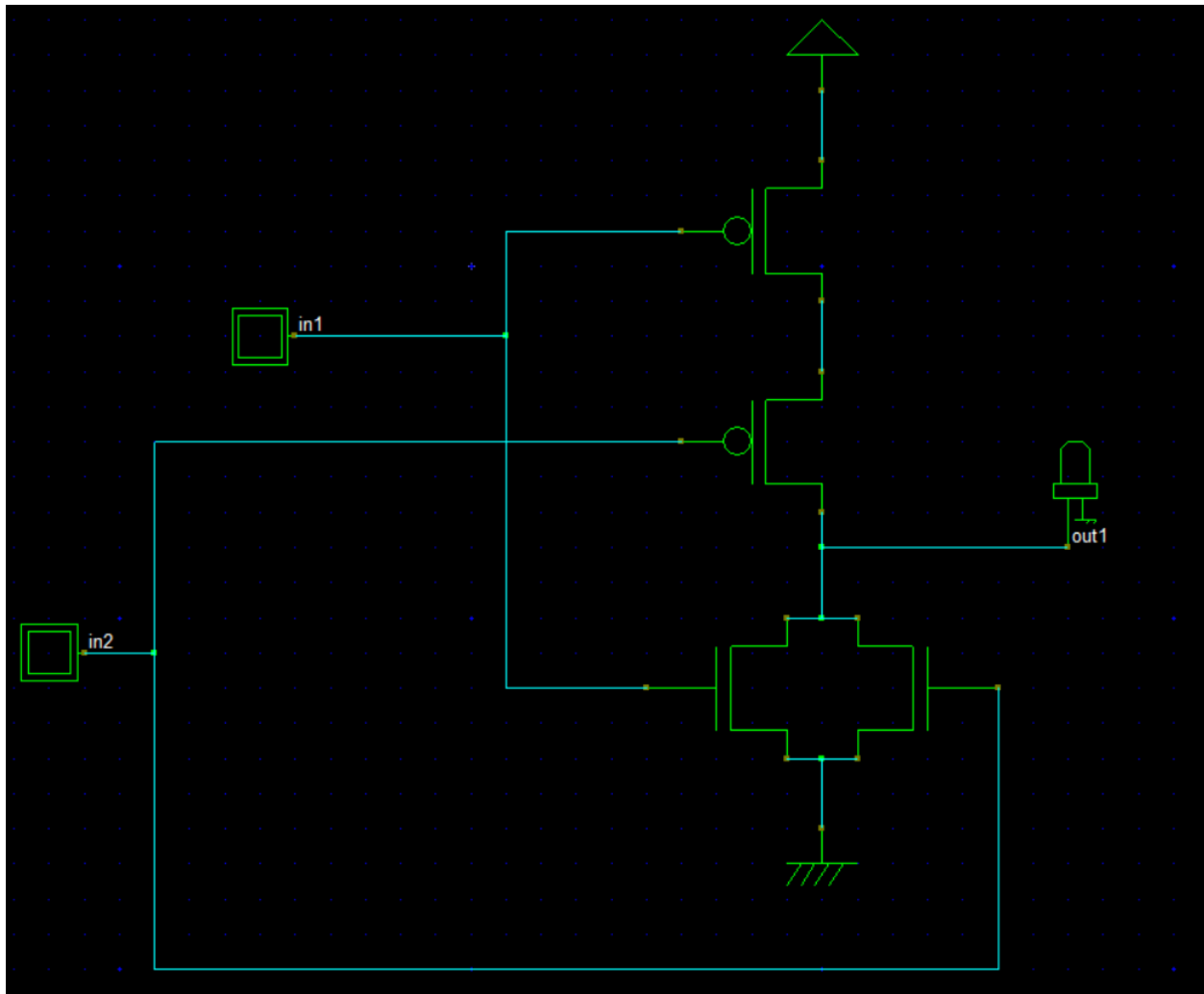


Timing Diagram -

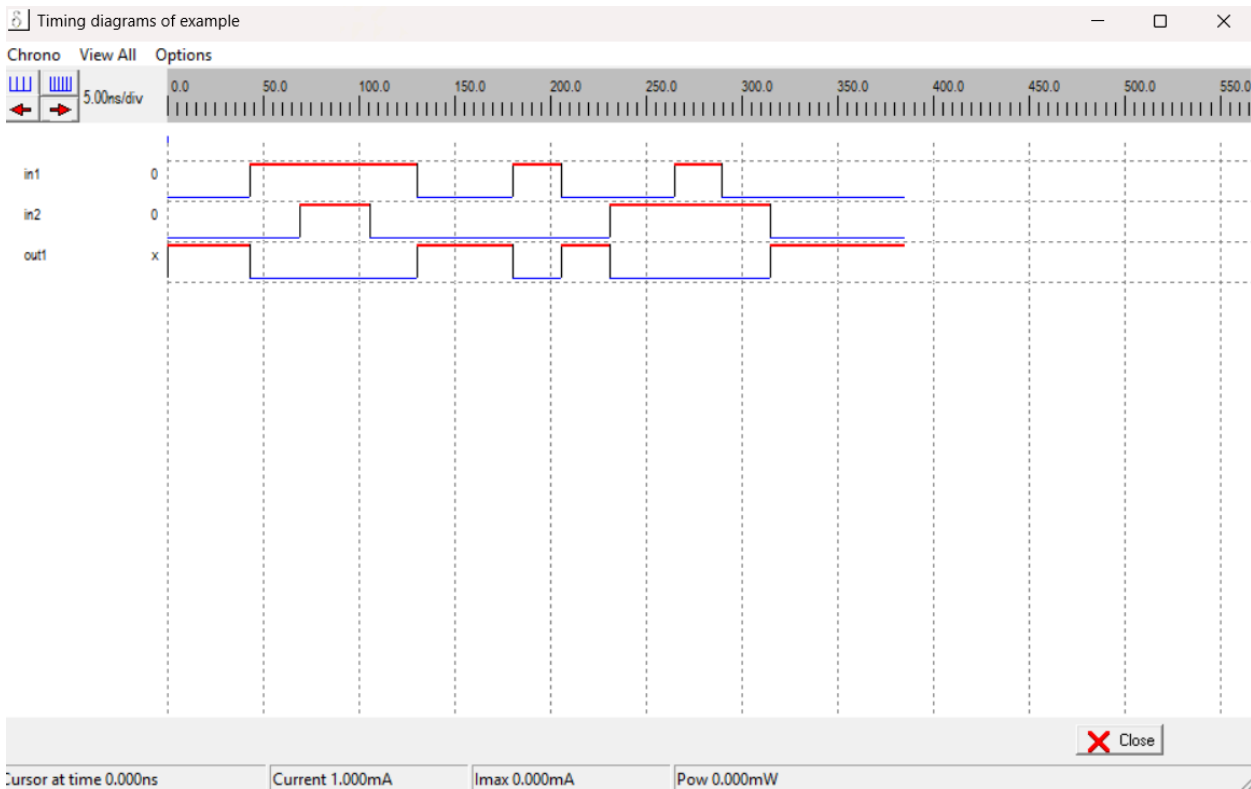


Circuit 03 - Nor Gate

Circuit -

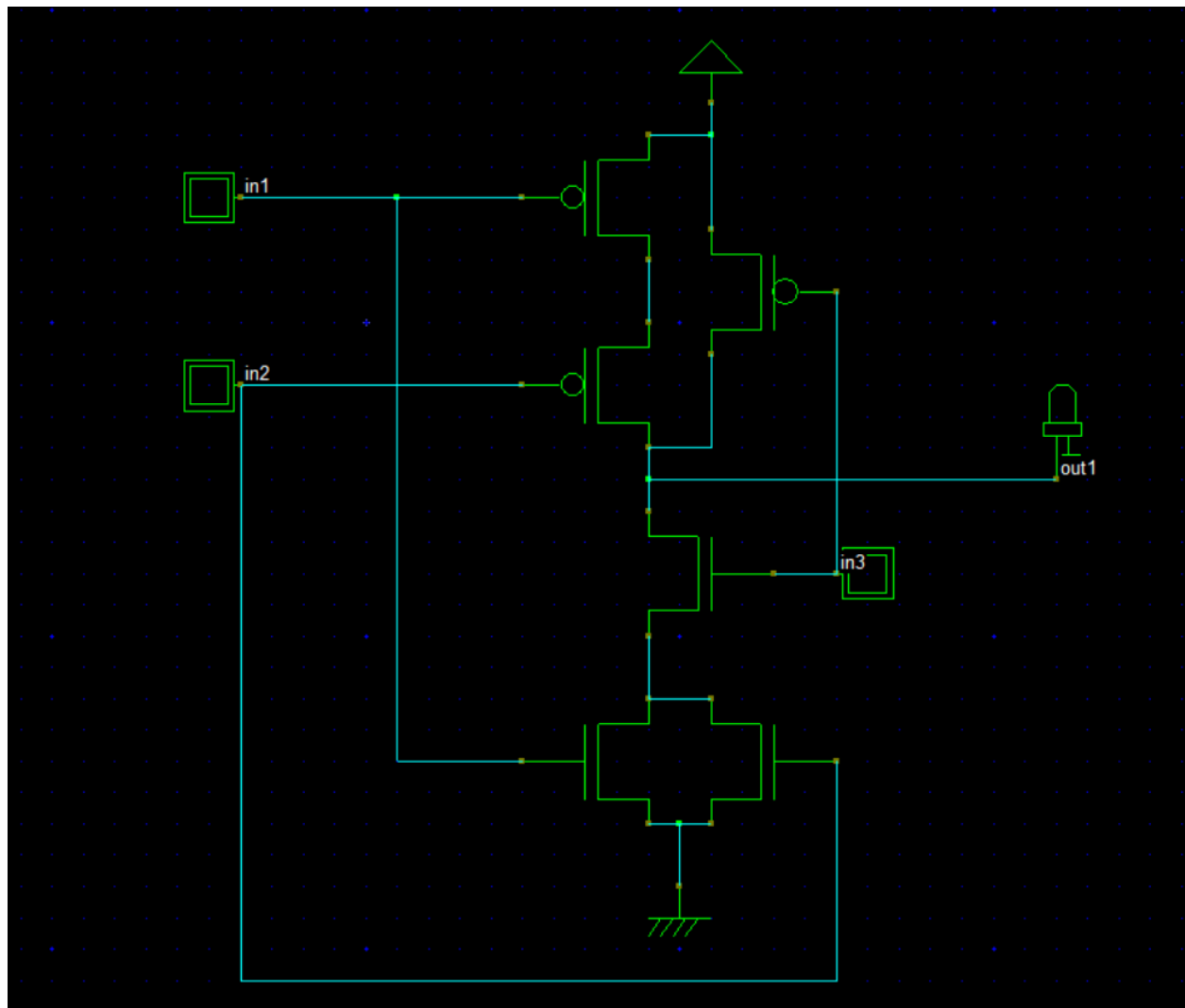


Timing Diagram -

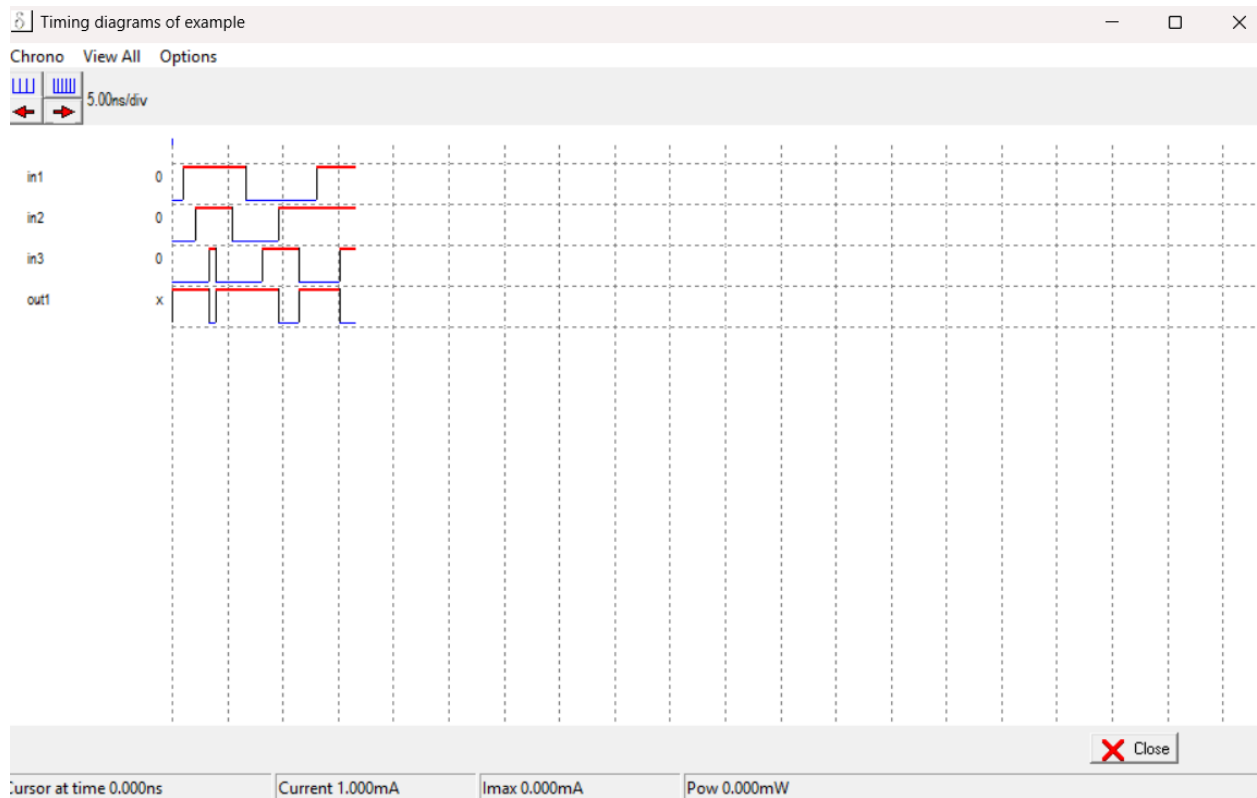


Circuit 04 - $Y = (A + B) \cdot C$ (whole bar)

Circuit -



Timing Diagram -



Conclusion

In this work, the inverter, NAND, NOR, and the logic function $Y = (A + B) \cdot C$ (whole bar) were successfully designed using Microwind Lite. The layouts were implemented correctly, and their corresponding time diagrams confirmed proper logical operation for all input combinations. The simulation results matched the theoretical truth tables, validating the functionality of each circuit.